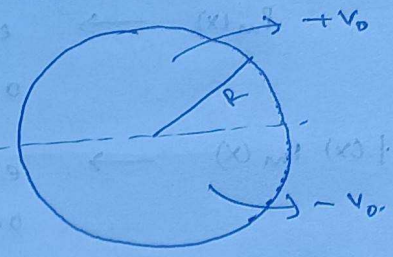
Tutorial - 1

(8) Calculate $V(\vec{r})$

(a) Inside the sphere

(b) Outside " "



Ans: For calculating this, we solve the Laplace's Eqn in spherical coordinates. Since there is no dependence on ϕ; $\cancel{\Rightarrow 0}$ $\frac{\partial V}{\partial \phi} = \cancel{\frac{\partial^2 V}{\partial \phi^2}}\ 0$

$$\Rightarrow \frac{\partial^2 V}{\partial \phi^2} = 0$$

Laplace's Eqn : $\nabla^2 V = 0$

Solution in spherical coordinates (from A3)

$$V(r,\theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}}\right) P_l(\cos\theta)$$

Inside Sphere

Now, potential inside the sphere (@ $r=0$) $\longrightarrow$ finite

$\therefore$ $\boxed{B_l = 0}$ $\cancel{\longrightarrow}$ ~~Inside the~~ s. — (1)

$$\therefore \boxed{V(r,\theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos\theta)} \longrightarrow \text{Inside the sphere.}$$

$$V(R,\theta) = \begin{cases} V_0 & ; \theta \in [0, \frac{\pi}{2}] \\ -V_0 & ; \theta \in [\frac{\pi}{2}, \pi] \end{cases}$$

$$= V_0 \operatorname{sgn}(\cos\theta) = f(\theta).$$

~~$\theta \in [0, \frac{\pi}{2}]$~~ @ $r = R$

$$\underbrace{\cancel{V_0 \operatorname{sgn}(\cos\theta) V_0}}_{f(\theta)} = \sum_{l=0}^{\infty} A_l R^l P_l(\cos\theta) \quad \text{(from (1))}$$

$$\Rightarrow \int_{-1}^{1} f(\overset{x}{\cancel{\theta}})\, P_m(\overset{x}{\cancel{\cos\theta}})\, dx = \sum_{l=0}^{\infty} A_l R^l \frac{2}{2m+1} \delta_{ml} \quad [\text{Taking } x = \cos\theta]$$

$$\Rightarrow \int_{-1}^{1} f(x)\, P_m(x)\, dx = \frac{2}{2m+1} A_m R^m \quad \left[\int_{-1}^{1} P_l(x)\, P_m(x)\, dx = \frac{2}{2m+1}\delta_{ml}\right]$$

Now, $f(x) = V_0 \, \mathrm{sgn}(x) \longrightarrow$ odd f^n

$P_m(x) \longrightarrow$ even if m = even
odd if m = odd.

$f(x) P_m(x) \longrightarrow$ even if m = odd $\longrightarrow$ integral survives,
odd " " = even $\longrightarrow$ integral = 0

Hence, only odd m terms survive.

Odd m:

$$A_m = \frac{2m+1}{2R^m} \int_{-1}^{1} f(x) \, P_m(x) \, dx = \frac{2m+1}{R^m} \int_0^1 f(x) P_m(x) \, dx$$

(even integrand)

$$= \cancel{-V_0 \int_{-1}^{0} P_m(x) \, dx} + \frac{V_0}{R^m} \int_0^1 P_m(x) \, dx \, (2m+1)$$

$$\cancel{= V_0 \left[-\int_{-1}^{0} P_m(x) \, dx + \int_0^1 P_m(x) \, dx \right]}.$$

$$\int_0^1 P_m(x) \, dx = \int_0^1 \frac{P'_{m+1} - P'_{m-1}}{(2m+1)} = \frac{1}{2m+1} \left(P_{m+1} - P_{m-1} \right) \quad \text{— (ii)}$$

$$\Rightarrow \boxed{\int_0^1 P_m(x) \, dx = \frac{1}{2m+1} \left(P_{m-1}(0) - P_{m+1}(0) \right)} \quad [\because P_L(1) = 1] \quad \text{— (iii)}$$

$\cancel{= 0}$ $\quad \cancel{= P_L}$

$$\cancel{\therefore \int_{-1}^{0} P_m(x) \, dx = \frac{1}{2m+1} \left[\left(P_{m+1}(0) - P_{m-1}(0) \right) - \left(P_{m+1}(-1) - P_{m-1}(-1) \right) \right]}$$

(from (ii))

$\cancel{L \rightarrow \text{odd} \Rightarrow L+1, \; L-1 \rightarrow \text{even}}$

$\cancel{P_L(-x) = (-1)^L P_L(x) \Rightarrow P_L(-x) = P_L(x)}$ for even L.

$\cancel{\Rightarrow P_L(-1) = P_L(1) = 1.}$

$$\cancel{\therefore \int_{-1}^{0} P_m(x) \, dx = \frac{1}{2m+1} \left(P_{m+1}(0) - P_{m-1}(0) \right)} \quad \text{— (iv)}$$

$$\cancel{\therefore \text{(iii)} + \text{(iv)}, \quad \frac{1}{2m+1} \left[P_{m-1}(0) - P_{m+1}(0) - P_{m+1}(0) + P_{m-1}(0) \right]}$$

$$\cancel{= \frac{2}{2m+1} \left[P_{m-1}(0) - P_{m+1}(0) \right]}$$

$$\therefore A_m = \frac{\cancel{(2m+1)} V_0}{R^m} \cdot \frac{1}{\cancel{2m+1}} \left[P_{m-1}(0) - P_{m+1}(0) \right]$$

~~Now,~~ $\cancel{(m+1) P_{m+1}(x) - (2m+1) \times P_m(x) + m P_{m-1}(x) = 0}$

~~@ $x = 0$:~~

$$\cancel{(m+1) P_{m+1}(0) + m P_{m-1}(0) = 0}$$

$$\therefore \quad A_m = \begin{cases} \dfrac{V_0}{R^m}\left[P_{m-1}(0) - P_{m+1}(0)\right] ; & m = \text{odd} \\ 0 \quad ; & m = \text{even} \end{cases}$$

$$\therefore \quad \boxed{V_{in}(r,\theta) = \sum_{\substack{m=1 \\ (\text{odd})}}^{\infty} V_0\left[P_{m-1}(0) - P_{m+1}(0)\right]\left(\frac{r}{R}\right)^m P_m(\cos\theta)}$$

(Ans)

Outside Sphere :

Tutorial :

As $r \to \infty$; $V \to 0$

$$\therefore \quad \boxed{A_L = 0}$$

$$\therefore \quad \boxed{V(r,\theta) = \sum_{l=0}^{\infty} \frac{B_L}{r^{L+1}} P_L(\cos\theta)}$$

@ $r = R$:

$$f(x) = \sum_{l=0}^{\infty} \frac{B_L}{R^{L+1}} P_L(x)$$

$$\Rightarrow \int_{-1}^{1} f(x) P_m(x) = \frac{B_m}{R^{m+1}} \frac{2}{2m+1}$$

$$\Rightarrow \boxed{B_m = \frac{(2m+1)R^{m+1}}{2} \int_{-1}^{1} f(x) P_m(x)\, dx}$$

Using the same logic as before, the integral only survives for $m = \text{odd}$.

Odd m

$$B_m = \frac{2m+1}{R^{(m+1)}} \cdot \int_0^1 f(x) P_m(x)\, dx \quad (\text{even integral})$$

$$= \frac{(2m+1)}{R^{(m+1)}} V_0 \cdot \frac{1}{2m+1} \left[P_{m-1}(0) - P_{m+1}(0)\right]$$

(from (iii))

$$B_m = \frac{V_0}{R^{(m+1)}} \left(\quad \right) ; \quad m = \text{odd}$$

$$\Rightarrow B_m = 0 \quad ; \quad m = \text{even}.$$

$$V_{out}(r,\theta) = \sum_{\substack{m=1 \\ (odd)}}^{\infty} \cancel{V_0 \left(\frac{r}{R}\right)} \; V_0 \left(\frac{R}{r}\right)^{m+1} \left[P_{m+1-1}(0) - P_{m+1}(0) \right] P_m(\cos\theta)$$

(Ans)